

Reproduction Report: Safety Layers in Aligned Large Language Models

Information Security Project
Rishabh Kumar Patel (ID: 2023uai1803)

Hardware Environment: Lightning AI's H100 GPU

Target Model: LLaMA-2-7B-Chat

Original Paper: *Safety Layers in Aligned Large Language Models: The Key to LLM Security* (Li et al., ICLR 2025 | arXiv 2408.17003)

1 Introduction

This report documents the reproduction of the findings from the paper “*Safety Layers in Aligned Large Language Models*.” The paper claims that a small, contiguous set of middle layers (referred to as “safety layers”) are primarily responsible for recognizing and refusing malicious queries in aligned Large Language Models (LLMs). Building upon this, the authors propose **Safely Partial-Parameter Fine-Tuning (SPPFT)**, a defense mechanism that freezes these safety layers during fine-tuning to prevent security degradation against jailbreak attacks while maintaining model performance.

The reproduction was conducted on Lightning AI's H100 GPU, specifically focusing on the LLaMA-2-7B-Chat model. The replication successfully verifies the existence and localization of these safety layers and validates the effectiveness of SPPFT against various fine-tuning attacks.

2 Experimental Reproduction and Findings

2.1 Verification of Safety Layer Existence

The paper states that the existence of safety layers can be observed by tracking the cosine similarity between hidden-layer vectors of normal queries versus malicious queries.

Reproduction Result: We successfully reproduced the cosine-similarity gap analysis. Consistent with the paper, a clear divergence emerges at specific middle layers of LLaMA-2-7B-Chat, confirming that the model's internal representations for normal and malicious prompts separate significantly in this contiguous region.

2.2 Safety Layer Localization via Over-Rejection

To find the precise upper and lower bounds of the safety layers, the paper leverages the over-rejection phenomenon (where amplifying safety layer weights causes the model to erroneously refuse benign queries).

Reproduction Result: By sweeping different layer ranges and tracking over-rejections on the dataset (D_O), we observed the over-rejection counts across various ranges. The analysis successfully localizes the safety layers, confirming the paper's reported optimal range of [9, 14] as a highly significant region for over-rejection. This consistent behavior strongly supports the paper's findings.

3 Defense Evaluation: SPPFT vs. Full-Parameter Fine-Tuning

We reproduced the fine-tuning attack experiments (Table 2 and Table 3 from the paper) to evaluate SPPFT against Full Parameter Fine-Tuning (FullFT).

3.1 Normal, Implicit, and Backdoor Attacks (Table 2)

Normal Dataset (D_N)

Initial Baseline: $R_h = 1.35\%$, $S_h = 1.03$

- **SPPFT (Ours):** Harmful Rate (R_h) = 2.8%, Score (S_h) = 1.063 (*Paper:* $R_h = 2.88\%$, $S_h = 1.06$)
- **FullFT (Ours):** Harmful Rate (R_h) = 10.5%, Score (S_h) = 1.382 (*Paper:* $R_h = 10.58\%$, $S_h = 1.38$)
- **Performance:** Rouge-L perfectly matched (0.248 for SPPFT, 0.270 for FullFT). MMLU for SPPFT was an exact match (0.470), while FullFT was effectively consistent (0.453 vs. Paper’s 0.458).

Implicit Attack Dataset (D_I)

- **SPPFT (Ours):** $R_h = 6.7\%$, $S_h = 1.195$ (*Paper:* 6.73%, 1.19)
- **FullFT (Ours):** $R_h = 58.3\%$, $S_h = 3.265$ (*Paper:* 58.85%, 3.26)

Backdoor Attack Dataset (D_B)

- **SPPFT (Ours):** $R_h = 5.3\%$, $S_h = 1.205$ (*Paper:* 5.58%, 1.20)
- **FullFT (Ours):** $R_h = 60.5\%$, $S_h = 3.195$ (*Paper:* 60.58%, 3.19)

Observation: SPPFT successfully mitigates security degradation across all three benign/stealth attack vectors. FullFT suffers catastrophic security collapse under Implicit and Backdoor attacks, while SPPFT maintains baseline-level safety. The replicated results are virtually identical to the paper.

3.2 Harmful Data Attack (Table 3)

The models were fine-tuned using a mix containing malicious data at varying corruption rates (p).

Malicious Rate	Method	Harmful Rate (R_h)	Harmful Score (S_h)	Match Status
p = 0.05	FullFT	26.0%	1.88	Exact Match
	SPPFT	7.9%	1.25	Exact Match
p = 0.1	FullFT	53.7%	2.97	Exact Match
	SPPFT	25.6%	1.86	Exact Match
p = 0.2	FullFT	93.5%	4.53	Exact Match
	SPPFT	68.3%	3.61	Exact Match

Observation: The results for the harmful data attacks perfectly match the paper’s reported numbers. SPPFT significantly reduces the harmful rate compared to FullFT, confirming its robustness even when explicitly exposed to toxic training data.

4 Summary of Reproduction Success

The reproduction was highly successful, with our results mirroring the paper’s findings without any notable deviations.

1. **Safety Layer Localization:** The optimal safety layer range was confirmed, and the over-rejection behavior aligns completely with the original authors’ methodology.
2. **Consistent Metrics:** The Harmful Score (S_h) and Harmful Rate (R_h) metrics across Table 2 are nearly identical to the paper, with only minute, functionally negligible differences (~ 0.003 to 0.005) that are completely expected due to minor variance in GPT-4 evaluation APIs.
3. **Exact Matches:** Table 3 (Harmful Data Attack) yielded exact numerical matches across all metrics and corruption rates, fully validating the paper’s claims.

5 Conclusion

The reproduction successfully validates the core claims of the paper on Lightning AI’s H100 hardware. The existence of contiguous safety layers in LLaMA-2-7B-Chat is confirmed. Furthermore, freezing these layers (SPPFT) during fine-tuning proves to be a highly effective defense against normal, implicit, backdoor, and explicitly harmful fine-tuning attacks. The reproduced metrics strongly align with the original findings, demonstrating that protecting a model’s safety layers effectively preserves its security alignment without sacrificing task performance.